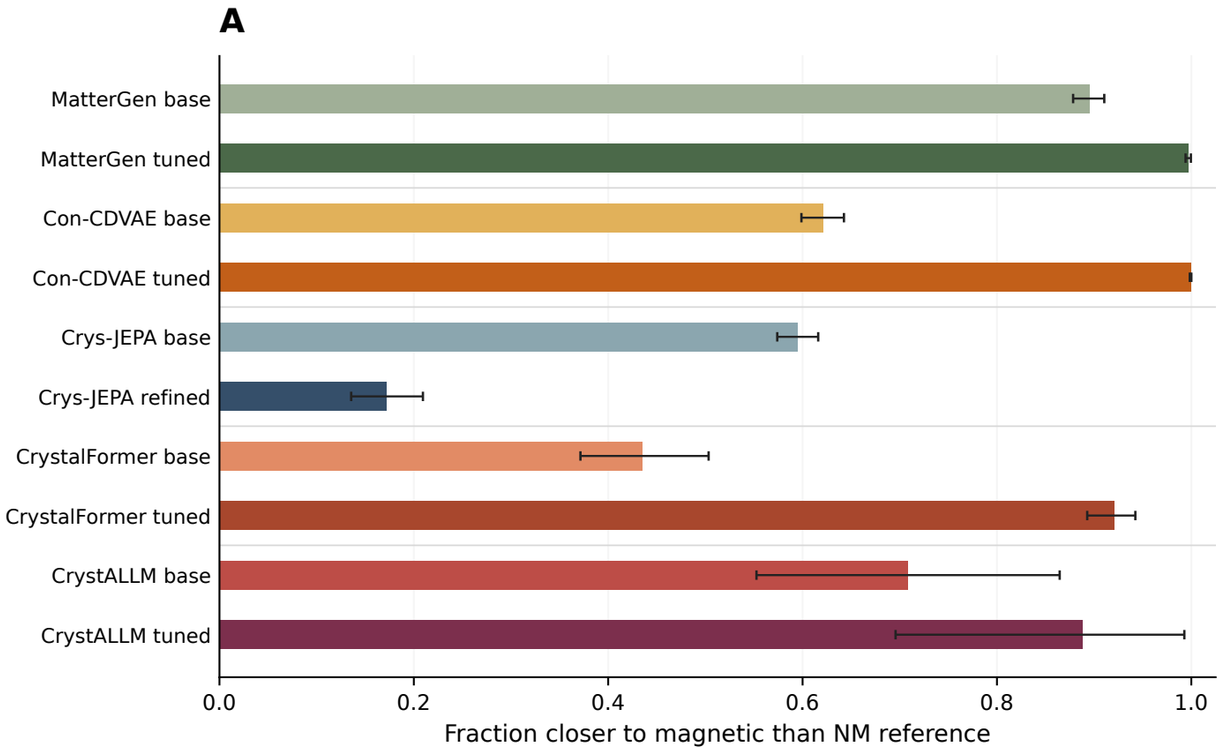
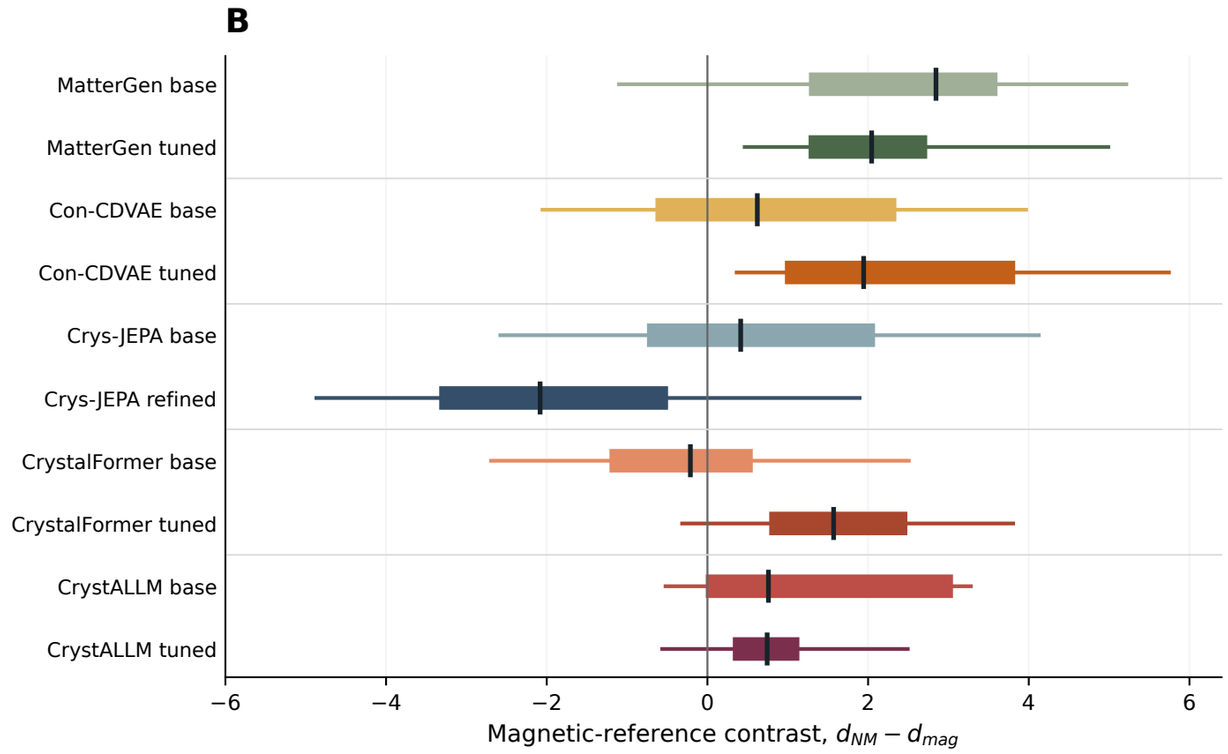
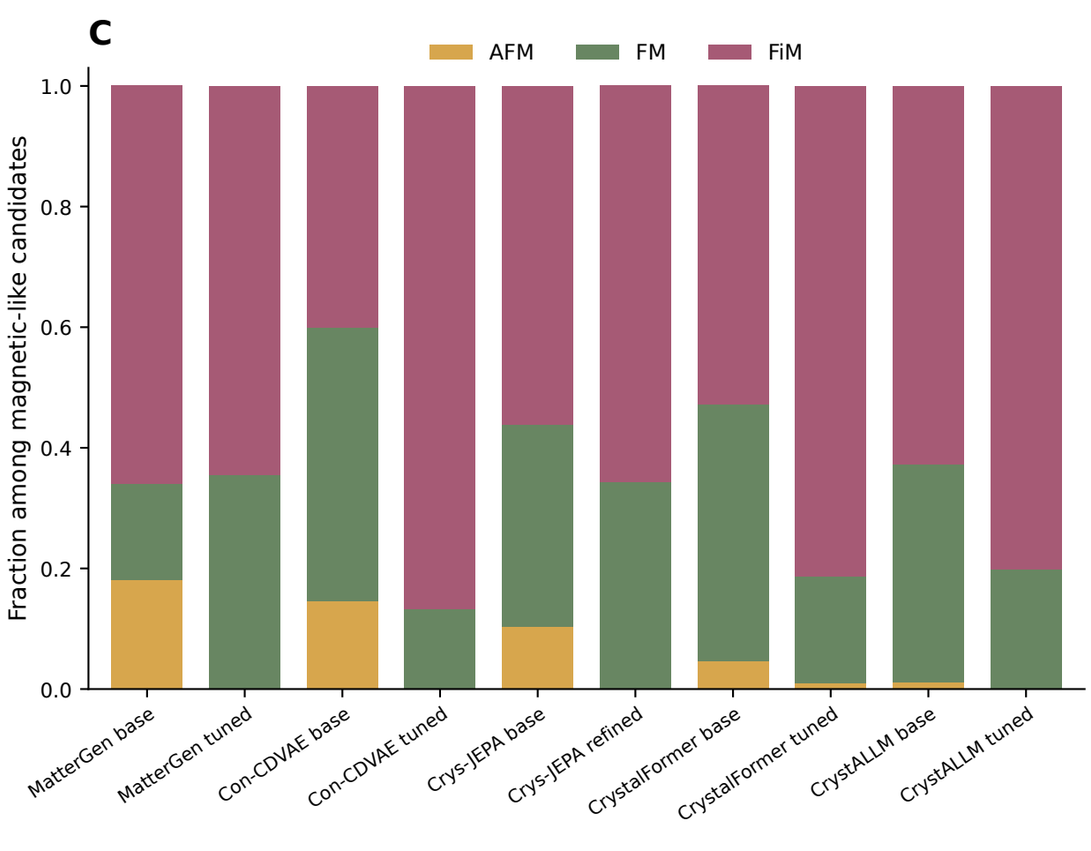
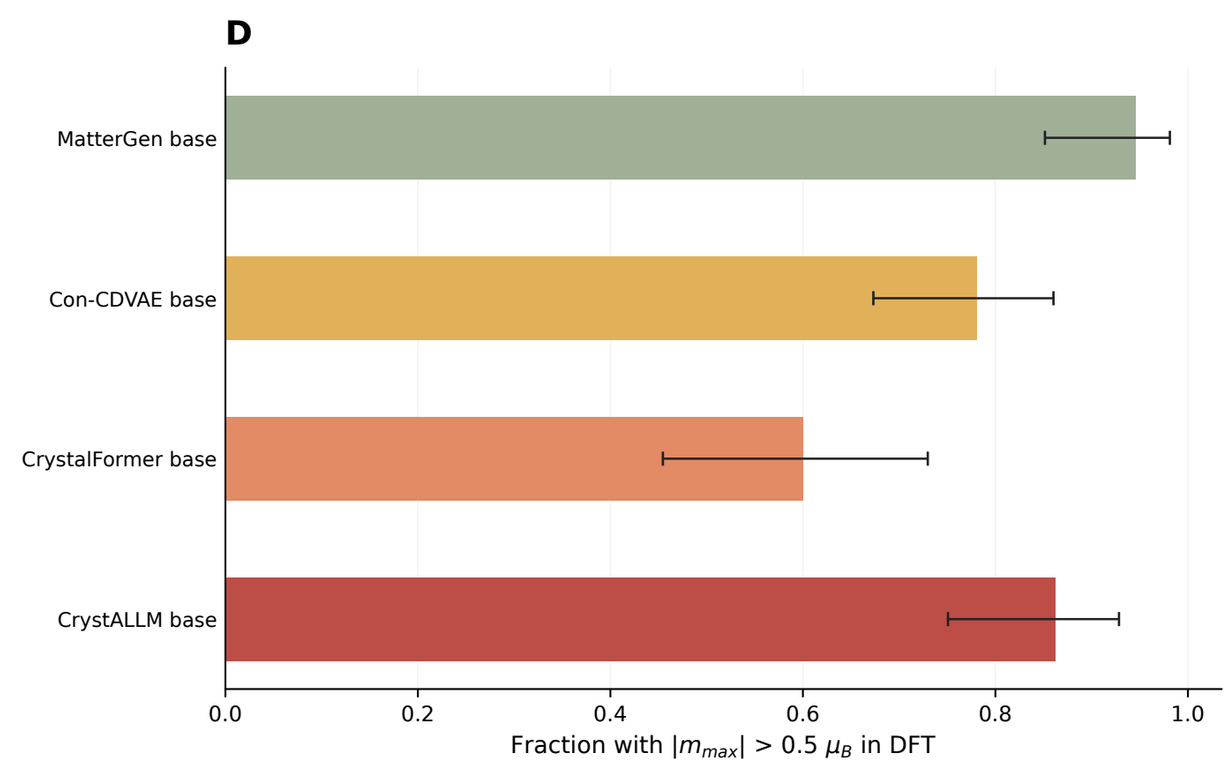

A
MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned
0.0
0.2
0.4
0.6
0.8
1.0
Fraction closer to magnetic than NM reference


B
MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned
−6
−4
−2
0
2
4
6
Magnetic-reference contrast, $d_{NM} - d_{mag}$


C
AFM
FM
FiM
Fraction among magnetic-like candidates
0.0
0.2
0.4
0.6
0.8
1.0
MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned


D
MatterGen base
Con-CDVAE base
CrystalFormer base
CrystALLM base
0.0
0.2
0.4
0.6
0.8
1.0
Fraction with $|m_{max}| > 0.5\ \mu_B$ in DFT